K Nikhil Reddy

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Professional Summary

- Experience in building robust backend and full-stack applications using NodeJS, React, Spring Boot, Kafka, Java and cloud-native with a strong track record of delivering high-impact solutions in fast-paced environments.
- Proven ability to collaborate cross-functionally with QA, DevOps, and UI/UX teams, combined with strong problem-solving and analytical thinking skills to architect scalable microservices, secure APIs, and optimize system performance.
- Highly adaptable and continuously learning engineer, with a passion for mastering new technologies and implementing modern development practices such as CI/CD, containerization, security protocols, and graph-based algorithmic logic.

TECHNICAL SKILLS:

- **Languages:** Java 8/11/17, Python, JavaScript, TypeScript, SQL, Shell Scripting, Bash, HTML, CSS
- **Backend Frameworks:** Spring Boot, Spring MVC, Spring Security, NestJS, Express.JS
- **Frontend:** Angular 12+, Next.js, React, Tailwind CSS, Bootstrap, jQuery, JSP
- **J2EE Technologies:** Servlets, JSP, JDBC, JUnit, JAXB, JMS, JSTL, Multi-threading, EJB, JavaBeans
- **Databases:** MySQL, PostgreSQL, Oracle DB, MongoDB, Redis
- **APIs & Integration:** REST, JSON, Swagger, Postman, OAuth2, JWT, SOAP, GraphQL
- **Messaging & Eventing:** Apache Kafka, RabbitMQ
- **DevOps Tools:** Jenkins, GitHub Actions, Maven, Gradle, Docker, Kubernetes
- **Cloud Platforms:** AWS (EKS, EC2, RDS), Azure (App Services, DevOps)
- **Version Control & CI/CD:** Git, GitLab, Bitbucket, Jenkins, GitHub Actions
- **Testing Frameworks:** JUnit5, TestNG, Mockito, Selenium, REST Assured
- **Data Science & other tools:** Data Structures & Algorithms, Scikit-learn, TensorFlow, Numpy, Pandas
- **Monitoring & Logging:** ELK Stack, Prometheus, Grafana, Splunk, Log4j
- **Project Management:** JIRA, Confluence, Agile/Scrum

EXPERIENCE

Client: ECP, USA **Jan 2024 – Present**

Role: Software Developer

- Built the front-end architecture of the corporate training platform from the ground up using **React/Next.js**, **TypeScript** and **Tailwind CSS** in VS Code, leveraging **Git** for version control to deliver dynamic, responsive and high-performance interfaces optimized for accessibility.
- Partnered closely with **UI/UX designers** to convert high-fidelity **Figma** designs into pixel-perfect, production-ready components, ensuring visual consistency, intuitive user flows, and seamless cross-browser compatibility across devices.
- Integrated interactive animations and **AOT** (Ahead-of-Time) compilation techniques to enhance page transitions, improve load times and optimize pages for **SEO** and Core Web Vitals metrics.
- Developed backend **APIs** using **NestJS** for authentication, course management and user dashboards, ensuring secure and efficient data exchange with the React/Next.js frontend.
- Deployed **Authorize.net** payment terminal using **NestJS** to enable secure, tokenized transactions and real-time subscription payment updates.
- Integrated the platform with **Absorb LMS** via RESTful APIs to synchronize course data and user progress while optimizing data fetching, pagination, and course recommendation logic in **Next.js** using **search** and **sorting algorithms** for improved performance and personalization.
- Integrated an **AI chatbot** using **Claude API** and **NestJS microservice** to deliver prediction-based, context aware **assistance** driven by **customer feedback** within the ECP platform.
- Collaborated with enterprise clients to customize and **integrate B2B solutions**, aligning backend APIs and data flows with specific business requirements and **LMS** systems
- Designed and implemented unit test suites in **Jest** to validate UI components and business logic, integrating them with **CI/CD pipelines** to ensure automated quality assurance and reliable releases.

Client: Cognizant, India **Feb 2022 - June 2023**

Role: Developer

- Refactored appointment scheduling logic using time slot **bucketing** and **hash-based indexing** on patient records, which reduced database scan time and improved query performance by 40% in high-volume booking modules.

- Developed and maintained **Spring Boot microservices** for managing electronic medical records (EMRs), securing endpoints with **JWT** and integrating with external systems via **REST** and **GraphQL** APIs for real-time patient data exchange.
- Optimized **Oracle DB** performance for medication and trial data by leveraging **PL/SQL** packages, stored procedures, and triggers with indexed views, improving API response times **by 30%** during clinical decision-making workflows.
- Implemented caching of frequently accessed patient appointment data using **Redis**, reducing database load and improving API response times by 30%, while storing unstructured medical notes and audit logs efficiently in **MongoDB** for flexible querying and faster retrieval
- Automated **CI/CD** workflows using **Cloud Build** and **Docker**, enabling smooth deployment of containerized microservices to Google Cloud Run and **GKE**, reducing deployment rollback time by **60%** during hotfixes.
- Automated routine system administration tasks on Windows environments using **PowerShell scripts**, improving deployment efficiency and reducing manual errors by 35%.
- Developed responsive **Next.js (React.js)** interfaces with **Tailwind CSS** & **TypeScript** for lab report uploads and prescription management, improving interaction times by 25% in low-bandwidth hospital environments.
- Improved patient portal with dynamic validations and role-based dashboards in **React.js**, boosting self-service adoption by 40% and reducing appointment errors.
- Monitored application logs and patient data access with **ELK stack** and **Splunk**, detecting and resolving critical issues to maintain 99.9% system uptime and ensure smooth healthcare operations.
- Coordinated **sprint** planning and task tracking through Jira dashboards, enabling early identification of blockers and improving feature delivery predictability by 20%.
- Followed **Agile methodologies** in two-week sprints, working closely with **QA, UI/UX**, and product teams to deliver backend features on time and incorporate continuous feedback.

Client: Adobe, India **Jan 2020 – Feb 2022**
Role: Associate Java Developer

- Developed **Java** backend services for managing digital media content by creating **modular classes** and interfaces, which improved code reuse and reduced maintenance time by 30%.
- Leveraged **Servlets** and **JSP** to implement dynamic content rendering and user session management, enhancing responsiveness and user interaction within the media management system.
- Enhanced **content search** and **sorting** by implementing optimized algorithms within service methods, which cut response times by 25%, improving user navigation speed.
- Increased software reliability by practicing Test-Driven Development (**TDD**), writing **unit tests** with **JUnit4** and **Mockito** before implementing features, which raised test coverage to 85% & reduced post-release bugs by 40%.
- Improved API robustness by implementing centralized **error handling** that returned consistent error messages, reducing client-side error troubleshooting time significantly.
- Optimized **MySQL** database schemas by adding indexes and restructuring tables, which reduced query execution time by 35% and sped up content retrieval processes.
- Created **Bash scripts** to automate routine maintenance tasks on Linux servers supporting media services, which reduced manual errors during deployments and improved system uptime.

CERTIFICATIONS

AWS Certified Developer Associate - DVA-C02

EDUCATION

Masters in Information Systems - Saint Louis University (Dec 2024)